Light

# GCSE Computer Science — Algorithms: Searching and Sorting (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A programmer needs to search for a value in a list. The search starts at position 0 and checks each position in turn until the value is found. What type of search is being described?

- ○ **A.** Binary search
- ○ **B.** Merge search
- ○ **C.** Linear search
- ○ **D.** Bubble search

Q2 (1 mark)

A student wants to use binary search on a list of student names. What must be true about the list before binary search can be used?

- ○ **A.** The list must be in alphabetical order
- ○ **B.** The list must contain fewer than 100 items
- ○ **C.** The list must contain only unique names
- ○ **D.** The list must be stored in an array, not a list

Q3 (1 mark)

During bubble sort, two adjacent items are compared and found to be in the wrong order.
What happens next?

- ○ **A.** The entire list is reversed
- ○ **B.** The smaller item is deleted
- ○ **C.** The algorithm restarts from the beginning
- ○ **D.** The two items swap positions

Q4 (1 mark)

Merge sort uses a 'divide and conquer' approach.
What does 'divide' mean in the context of merge sort?

- ○ **A.** Splitting the list into smaller sublists
- ○ **B.** Dividing each number by 2
- ○ **C.** Removing duplicate values
- ○ **D.** Calculating the average of the list

Q5 (1 mark)

A linear search is looking for the value 42 in a list.
Under what condition does the search stop?

- ○ **A.** When half the list has been checked
- ○ **B.** When the first item is checked
- ○ **C.** When 42 is found or the entire list has been checked
- ○ **D.** When a value greater than 42 is found

Q6 (1 mark)

In binary search, the algorithm first examines the middle element of the list.
Why is the middle element checked first?

- ○ **A.** Because middle elements are usually the target
- ○ **B.** Because the middle is easiest to access
- ○ **C.** Because smaller lists are always sorted from the middle
- ○ **D.** Because it allows half the list to be eliminated based on the comparison

Q7 (1 mark)

A list has 7 items. During one complete pass of bubble sort, how many comparisons are made?

- ○ **A.** 7
- ○ **B.** 14

- **C.** 49
- **D.** 6

Q8 (1 mark)

In merge sort, two sorted sublists need to be combined into one sorted list.
What is this process called?

- **A.** Dividing
- **B.** Swapping
- **C.** Merging
- **D.** Partitioning

Q9 (1 mark)

A list contains 1000 items. In the best case, how many comparisons does linear search need to find a target?

- **A.** 1
- **B.** 10
- **C.** 500
- **D.** 1000

Q10 (1 mark)

A sorted list contains 64 items. After two comparisons in binary search, what is the maximum number of items that could remain in the search

area?

- A. 64
- B. 32
- C. 16
- D. 8

Q11 (1 mark)

A programmer implements bubble sort with an optimisation that stops early if the list is already sorted.
How does the algorithm detect that the list is sorted?

- A. By checking if the first element is smallest
- B. By comparing the list to a backup copy
- C. By completing a pass with zero swaps
- D. By counting the total number of elements

Q12 (1 mark)

During the divide phase of merge sort, why does the algorithm stop dividing when sublists contain one element?

- A. Because one element cannot be split further
- B. Because the computer runs out of memory
- C. Because one element lists are faster to search

- **D.** Because single elements are already sorted by definition

Q13 (2 marks)

The list [8, 3, 12, 5, 9, 1] is searched using linear search to find the value 5.
How many comparisons are needed to find it?

- **A.** 2
- **B.** 3
- **C.** 4
- **D.** 6

Q14 (2 marks)

A binary search is performed on [10, 20, 30, 40, 50, 60, 70] to find 60.
Which element is compared first?

- **A.** 40
- **B.** 10
- **C.** 60
- **D.** 70

Q15 (2 marks)

The list [4, 7, 2, 9] is being sorted using bubble sort.
What is the state of the list after the first complete pass?

- ○ **A.** [2, 4, 7, 9]
- ○ **B.** [4, 7, 2, 9]
- ○ **C.** [4, 2, 9, 7]
- ○ **D.** [4, 2, 7, 9]

Q16 (2 marks)

The list [6, 1, 8, 3, 2, 5] is being sorted using merge sort.
After the first two divisions, what sublists exist?

- ○ **A.** [6, 1, 8] and [3, 2, 5]
- ○ **B.** [6, 1] and [8] and [3, 2] and [5]
- ○ **C.** [6] and [1] and [8] and [3] and [2] and [5]
- ○ **D.** [1, 6, 8] and [2, 3, 5]

Q17 (2 marks)

A sorted list contains 128 items. What is the maximum number of comparisons binary search needs to find any item or confirm it is absent?

- ○ **A.** 7
- ○ **B.** 8
- ○ **C.** 64
- ○ **D.** 128

Q18 (2 marks)

The list [3, 1, 4, 2] is sorted using bubble sort.
How many swaps occur during the first complete pass?

- ○ **A.** 1
- ○ **B.** 3
- ○ **C.** 4
- ○ **D.** 2

Q19 (2 marks)

Two sorted sublists [3, 8] and [1, 5] are being merged.
In what order are items added to the merged result?

- ○ **A.** 3, 8, 1, 5
- ○ **B.** 1, 3, 5, 8
- ○ **C.** 1, 5, 3, 8
- ○ **D.** 3, 1, 8, 5

Q20 (2 marks)

A list of 1000 sorted items needs to be searched repeatedly throughout the day.
Which search algorithm would be more efficient for this scenario?

- ○ **A.** Linear search, because the code is simpler
- ○ **B.** Binary search, because sorted data allows logarithmic time complexity

- ○ **C.** Linear search, because it works on any data
- ○ **D.** Neither — both perform identically on sorted data

Q21 (2 marks)

A list of 10 items is in reverse order (largest to smallest). Using bubble sort to sort it into ascending order, what is the minimum number of complete passes required?

- ○ **A.** 1
- ○ **B.** 10
- ○ **C.** 45
- ○ **D.** 9

Q22 (2 marks)

A programmer is choosing between bubble sort and merge sort for a system with very limited memory.
Which statement about memory usage is correct?

- ○ **A.** Bubble sort requires additional memory proportional to the list size
- ○ **B.** Merge sort uses constant additional memory regardless of list size
- ○ **C.** Bubble sort sorts in-place using minimal additional memory
- ○ **D.** Both algorithms use the same amount of additional memory

Q23 (2 marks)

A binary search is performed on [5, 10, 15, 20, 25] looking for the value 12.
How does the algorithm determine that 12 is not in the list?

- ○ **A.** By checking every element and finding no match
- ○ **B.** By noticing that 12 is not a multiple of 5
- ○ **C.** By reaching a point where the search range is empty
- ○ **D.** By finding that 12 is between two adjacent elements

Q24 (3 marks)

The list [5, 2, 4, 1] is sorted using bubble sort.
How many complete passes are needed to fully sort this list?

- ○ **A.** 2
- ○ **B.** 3
- ○ **C.** 4
- ○ **D.** 5

Q25 (3 marks)

A binary search is performed on [2, 5, 8, 11, 14, 17, 20, 23, 26] to find 17.
Which values are compared with 17 during the search, in order?

- ○ **A.** 14, 17
- ○ **B.** 14, 20, 17

- ○ **C.** 11, 17
- ○ **D.** 14, 23, 17

Q26 (3 marks)

The list [7, 3, 9, 2] is sorted using merge sort.
Which sequence shows the complete divide phase followed by the merge phase?

- ○ **A.** Divide: [7,3,9,2]→[7,3][9,2]→[7][3][9][2]. Merge: [3,7] [2,9]→[2,3,7,9]
- ○ **B.** Divide: [7,3,9,2]→[7][3][9][2]. Merge: [2,3,7,9]
- ○ **C.** Divide: [7,3,9,2]→[3,7][2,9]. Merge: [2,3,7,9]
- ○ **D.** Divide: [7,3,9,2]→[7,3][9,2]. Merge: [2,3,7,9]

Q27 (3 marks)

A games company stores 10,000 player usernames in alphabetical order. New players join occasionally, and usernames must be checked for uniqueness before registration.
Which searching approach is most appropriate?

- ○ **A.** Linear search, because new players are rare
- ○ **B.** Binary search, because the data is sorted and checks are frequent
- ○ **C.** Bubble sort followed by linear search for each check
- ○ **D.** Linear search, because it can handle any data order

Q28 (3 marks)

A student claims: 'Binary search is always better than linear search.'
Which response correctly evaluates this claim?

- ○ **A.** True — binary search is faster in all cases
- ○ **B.** False — linear search is always faster because it starts at the beginning
- ○ **C.** False — binary search requires sorted data, which may not be available or efficient to maintain
- ○ **D.** True — all professional software uses binary search exclusively

Q29 (3 marks)

A programmer needs to sort a list of 1,000,000 items on a system with limited RAM.
What is a potential drawback of using merge sort for this task?

- ○ **A.** Merge sort cannot handle lists larger than 10,000 items
- ○ **B.** Merge sort requires additional memory proportional to the list size, which may exceed available RAM
- ○ **C.** Merge sort only works on numerical data, not text
- ○ **D.** Merge sort becomes slower than bubble sort for large lists

Q30 (3 marks)

A teacher explains that both linear search and binary search find items in lists, but their efficiency differs greatly.
Which statement best explains why binary search is more efficient on large sorted lists?

- ○ **A.** Binary search uses less memory than linear search
- ○ **B.** Binary search eliminates half the remaining data with each comparison, giving logarithmic time complexity versus linear search's sequential approach
- ○ **C.** Binary search works on unsorted data more efficiently
- ○ **D.** Binary search makes fewer memory accesses because it reads items in pairs

Total: **55** marks • Time: **30 min**

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